

Multi-Decker Emissive Supramolecular Architectures Based on Shape-Complementary Ligands Pair

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Dye aggregates have attracted a great deal of attention due to their widespread applications in organic light-emitting devices, light-harvesting systems, etc. However, the strategies to precisely control chromophores with specific spatial arrangements still remain a great challenge. In this work, a series of double- and triple-decker supramolecular complexes are successfully constructed by coordination-driven self-assembly of carefully designed shape-complementary ligands, one claw-like tetraphenylethylene (TPE)-based host ligand and three tetratopic or ditopic guest ligands. The spatial configurations of these assemblies (one double-decker and three “S-shaped” or “X-shaped” triple-decker structures) depend on the angles of these TPE-derived ligands. Notably, the three triple-decker structures are geometric isomers. Furthermore, photophysical studies show that these complexes exhibit different ratios of radiative (k_r) and non-radiative (k_{nr}) rate constant due to the different spatial arrangements of TPE moieties. This study provides not only a unique strategy for the construction of multi-stacks with specific spatial arrangement, but also a promising platform for investigating the aggregation behavior of fluorescent chromophores.

donor-acceptor compounds and exhibit potential applications in electronic, photonic, and photovoltaic fields.^[6] To obtain an in-depth understanding of the photophysical properties of dye aggregates, extensive efforts have been devoted to the construction of covalently linked and well-defined multi-decker structures.^[7] Although the covalently linked foldamer approach can provide access to functional multi-stacks with particular lengths and well-defined arrangements, it was encumbered by challenging molecular design and the high synthetic costs.^[7] The construction of multi-decker structures via non-covalently supramolecular self-assembly provided another promising way to study the well-defined stacks.^[4] Several approaches have emerged to construct multilayer and folded structures, such as backbone-directed self-assembly,^[8] template-controlled self-assembly,^[9] etc.^[10] However, these strategies are almost based

1. Introduction

Light-emitting materials have attracted considerable attention and been widely explored over the past few decades, due to their widespread applications in light-emitting diodes,^[1] organic photovoltaics,^[2] light-harvesting systems,^[3] etc. Different aggregation behaviors of chromophores could seriously affect the final luminescence properties, such as H-, J-, and X-aggregation.^[3,4] Therefore, the self-organization of fluorescent molecules into well-defined structures plays a vital role in elucidating the emission mechanism of fluorescence and developing novel fluorescent molecules.^[5] Multi-decker structures have attracted considerable attention as the unique ordered arrays, which can facilitate the transfer of electrons in

on the π - π interactions of polycyclic aromatic hydrocarbons (PAH), which often lead to an ambiguous arrangement of the dye aggregates due to the limited directionality of dispersion forces.^[4,11] Furthermore, these strategies cannot be applied to the chromophores with small conjugation, weak π - π interactions, and twisted molecular configuration, such as tetraphenylethylene (TPE), triphenylamine, etc. Thus, a more sophisticated strategy with widely applicable for various chromophores needs to be created for constructing multi-decker architectures with particular spatial orientation of chromophores.

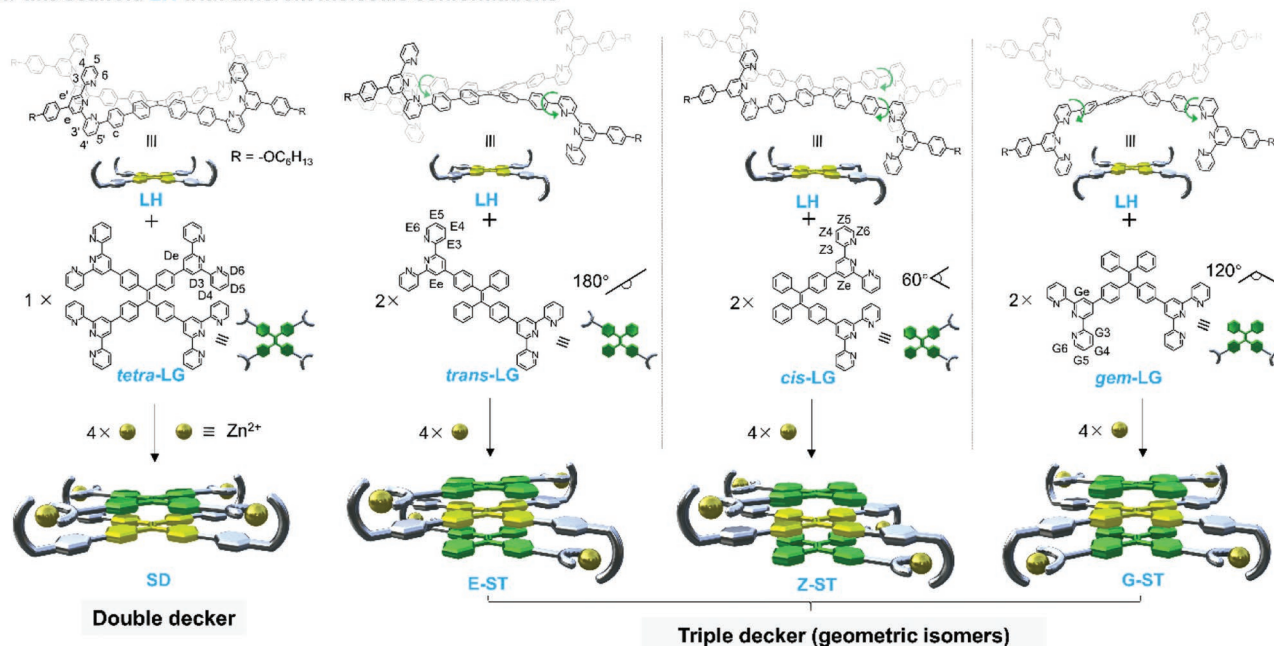
Inspired by the function and complexity of natural systems, the derivation of strategies to control the arrangement of different ligands to form multicomponent assemblies by integrative self-sorting has drawn much attention in recent years.^[12] Among them, the “shape-complementary” strategy has been proven to be a very powerful manner for constructing multicomponent assemblies in a non-statistical fashion rather than statistical mixtures,^[13] such as the heteroleptic cages assembled with pyridyl ligands and naked Pd(II).^[14] For various coordination groups in coordination-driven self-assembly, terpyridine (tpy) exhibited enhanced thermodynamic stability compared with pyridyl and bipyridyl ligands, and could overcome geometric changes during self-assembly owing to the higher number of coordination sites as well as higher bond energy.^[15] More importantly, the

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 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202202167>.

DOI: 10.1002/smll.202202167

claw-like scaffold LH with different molecule conformations



Scheme 1. Self-assembly of TPE-based double- and triple-decker supramolecular architectures. TPE moieties are indicated by different colors to distinguish the host and guest ligands.

tpy group has more modifiable sites at the pyridine ring on both sides, which provides more advantages in design and geometry for developing novel architectures. We hypothesized that the tpy-based shape-complementary ligands pair could provide a feasible and general approach to constructing multi-decker structures with a particular arrangement of chromophores.

In this study, a series of discrete double- and triple-decker structures constructed by shape-complementary strategy were reported (**Scheme 1**). The 6-position modification of tpy played a key role in the synthesis of host scaffold LH. Considering the significantly steric congestion caused by the ortho-modified structure, we designed TPE as the core of LH to realize the restriction of intramolecular rotation (RIR).^[16] Four guest ligands, including a tetratopic ligand tetra-LG and three ditopic ligands trans-LG, cis-LG, and gem-LG with geometric angles of 180°, 60°, and 120°, respectively, were carefully designed. After the self-assembly of host and guest ligands with Zn(II), double-decker [Zn₄(LH)(tetra-LG)] (SD) and triple-decker structures [Zn₄(LH)(trans-LG)₂] (E-ST), [Zn₄(LH)(cis-LG)₂] (Z-ST) and [Zn₄(LH)(gem-LG)₂] (G-ST) were successfully obtained, respectively. The spatial arrangements of TPE moieties were precisely controlled in the four supramolecular complexes, and their effect on photophysical properties in a dilute solution was investigated in detail to establish the structure-property relationship. Furthermore, most of the research on geometric isomers of TPE are limited to trans/cis isomers, however, the comparison of trans/cis/gem-configurations are very rare.^[17] In particular, there is no suitable supramolecular scaffold reported currently to precise control over the spatial orientation of trans/cis/gem isomers as far as we know.

2. Results and Discussion

2.1. Design, Synthesis, and Characterization of Ligands and Complexes

The detailed synthetic routes for TPE-containing ligands were shown in Supporting Information Scheme S1. Benefiting from the characteristic claw-like geometry, the 6-position modification of tpy enables the construction of a host for multi-decker structures via the shape-complementary principle. Claw-like motif 4 was efficiently obtained in 73% yield by the Kröhnke reaction of 2 and 3. The alkyl chain (R = -OC₆H₁₃) was introduced at the para-position of the peripheral phenyl group to improve the solubility of assembled complexes. After Suzuki coupling reactions of 5 and 6, host ligand LH was obtained as a green powder in 60% yield. Relatively, tetratopic ligand tetra-LG (yield 70%) and ditopic ligand gem-LG (yield 65%) were also obtained by the same reaction as ligand LH. However, for ditopic trans-LG and cis-LG, the separation of E/Z isomers of TPE with similar polarities via the McMurry coupling reaction was considerably challenging.^[18] To solve this problem, two approaches were developed to prepare trans-/cis-LG (**Scheme S1**, Supporting Information) in high yields (57% for trans-LG and 68% for cis-LG), and their structures were confirmed by single-crystal analysis (**Figure 1**, Tables S3 and S4, Supporting Information).^[19] The chemical structures of all of the above ligands were characterized by ¹H nuclear magnetic resonance (¹H NMR) spectroscopy, 2D-correlation spectroscopy (2D-COSY), and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry (Figures S5 and S53, Supporting Information). The double- and triple-decker complexes were assembled by mixing host ligand LH, guest

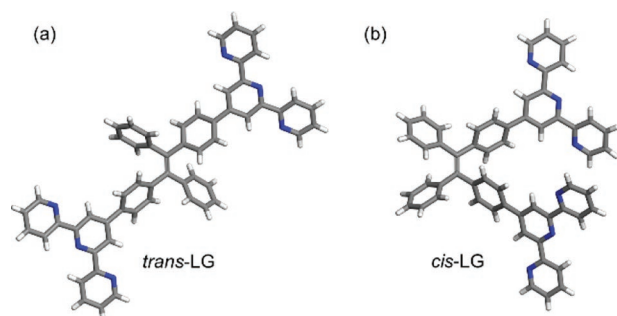


Figure 1. X-ray crystal structures of ligands a) *trans*-LG and b) *cis*-LG. (C, gray; H, white; N, blue).

ligands *tetra*-LG and *trans*/*cis*/*gem*-LG, and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a precise stoichiometric ratio ($\text{LH}:\text{tetra-LG}:\text{Zn}^{2+} = 1:1:4$, $\text{LH}:\text{trans}/\text{cis}/\text{gem-LG}:\text{Zn}^{2+} = 1:2:4$) in CHCl_3 and MeOH ($v/v = 1:2$) at 50°C for 12 h (Scheme 1). After that, excessive NH_4PF_6 was added to methanol to afford light-yellow precipitates in good yields (i.e., 91%, 88%, 90%, and 91% for SD, E-ST, Z-ST, and G-ST, respectively).

^1H NMR spectra of the ligands and complexes were shown in **Figure 2**. Owing to the modification on 6-position of *tpy*, three sets of signals for pyridine corresponding to host ligand LH were observed (Figure 2c,h). For the symmetrically modified four guest ligands *tetra*-LG and *trans*/*cis*/*gem*-LG, only two sets of signals for pyridine were observed, indicative of the successful preparation of these designed ligands (Figure 2a,e,f,j). Compared with those of ligands, the protons of the complexes exhibited remarkably broadened peaks, due to the slower tumbling motion of large complexes on the NMR time scale.^[20] All of the signals for these four complexes were accurately assigned by 2D correlation spectroscopy (COSY) and nuclear overhauser effect spectroscopy (NOESY) (Figures S32 and S53, Supporting Information). The proton signals corresponding to the methylene group ($-\text{OCH}_2-$) were observed at ≈ 4.50 ppm, and the integral ratios as expected for aromatic peaks were identified, suggestive of the probable formation of a single component. All proton signals corresponding to the 6-position of *tpy* exhibited upfield shifts, and the protons at 3- and 5-positions accordingly

exhibited downfield shifts, which were consistent with that reported previously.^[21] The split of the *tpy* signals of the guest ligands indicated that the rotation of *tpy* moieties is completely restricted in the altered chemical environment.^[16a] Notably, all H_c protons corresponding to the host ligand LH in SD, E-ST, Z-ST, and G-ST produced extremely large upfield shifts ($\Delta\delta \approx 1.60$ ppm for SD, $\Delta\delta \approx 1.71$ ppm for E-ST, $\Delta\delta \approx 1.68$ ppm for Z-ST, and $\Delta\delta \approx 1.77$ ppm for G-ST), indicative of a strong electronic shielding effect. In addition, compared with other assemblies, G-ST exhibited considerably sharper and well-resolved NMR signals, which was attributed to the smaller conjugated degree of ligand *gem*-LG. Diffusion-ordered NMR spectroscopy (DOSY) further provided dimensional information for these four complexes.^[22] All proton signals were observed at the same band for each complex, indicative of the formation of discrete assemblies (Figures S54 and S57, Supporting Information). Diffusion coefficients for SD, E-ST, Z-ST, and G-ST in CD_3CN were $\log D = -9.45$, $\log D = -9.48$, $\log D = -9.48$, and $\log D = -9.47$, respectively. The experimental hydrodynamic radii (r_H) of these complexes were calculated using the modified Stokes–Einstein equation (3.4 nm for SD, 3.6 nm for E-ST, 3.6 nm for Z-ST, and 3.5 nm for G-ST), which were consistent with the modeled structures.

Electrospray ionization-mass spectrometry (ESI-MS) and traveling wave ion mobility-mass spectrometry (TWIM-MS) (**Figure 3a–h**) were employed to further characterize these four complexes.^[23] In the ESI-MS spectrum of double-decker SD, one prominent set of peaks with different charge states ($3+$ to $7+$) was observed by the loss of PF_6^- counterions. The molecular weight of SD was 4937 Da, which was in good agreement with the expected chemical composition of one host ligand LH, one guest *tetra*-LG, four $\text{Zn}(\text{II})$ ions, and eight PF_6^- . Relatively, a set of dominant peaks with continuous charge states ranging from $3+$ to $6+$ was observed in the ESI-MS of geometric isomers E-ST, Z-ST, and G-ST, indicating that two guest ligands were present in the assembled complexes E-ST, Z-ST, and G-ST (Figure 3b–d). Further investigation revealed that all the experimental isotope patterns of these four complexes were in excellent agreement with the corresponding simulated isotope patterns (Figures S1 and S4, Supporting Information).

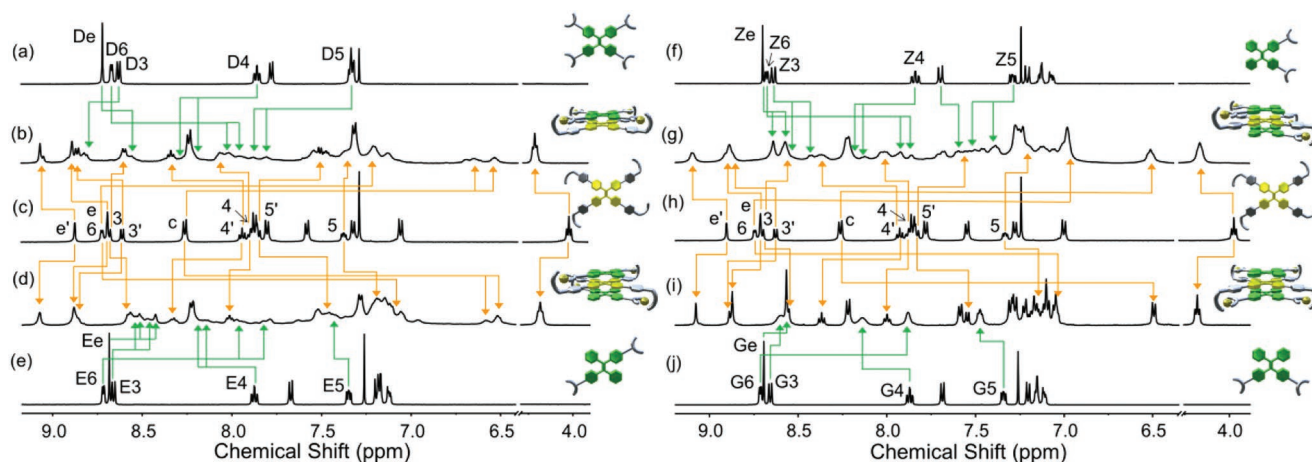


Figure 2. ^1H NMR spectra (500 MHz, 300 K) of ligands a) *tetra*-LG, c) and h) LH, e) *trans*-LG, f) *cis*-LG, and j) *gem*-LG in CDCl_3 and complexes b) SD, d) E-ST, g) Z-ST, and i) G-ST in CD_3CN .

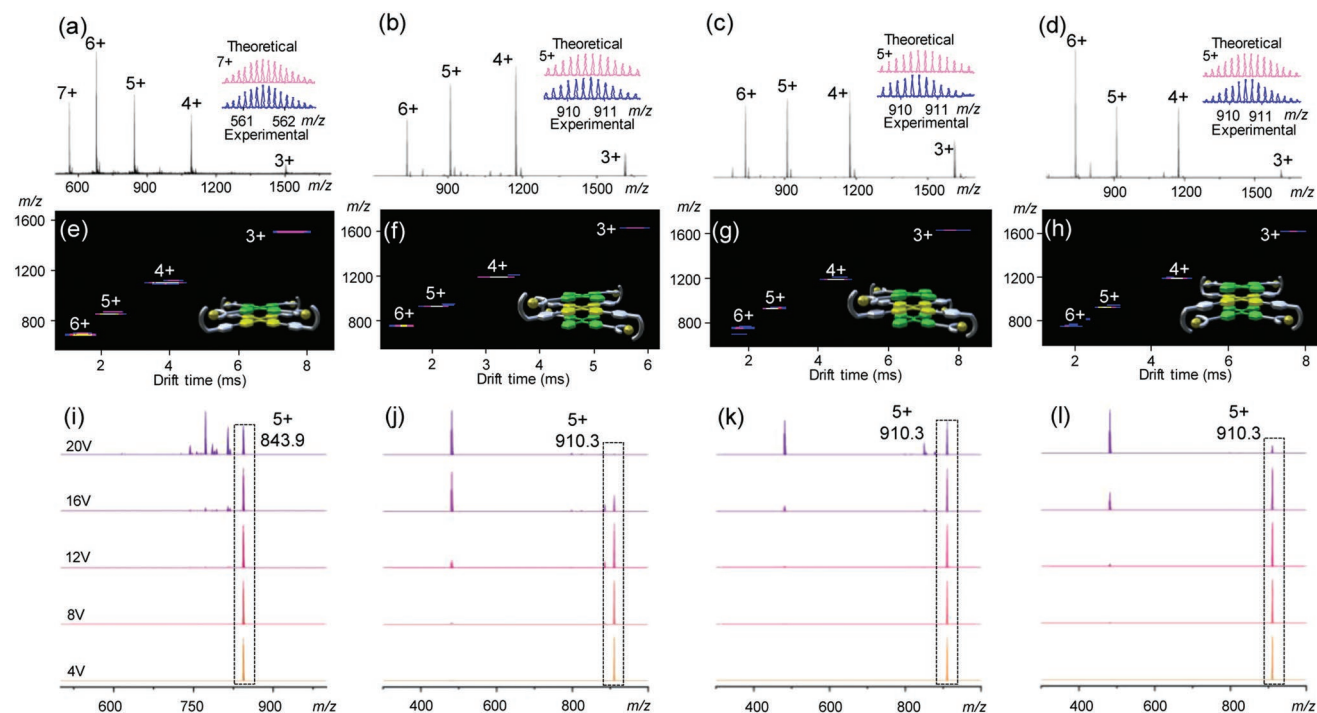


Figure 3. ESI-MS and TWIM-MS spectra of SD (a,e), E-ST (b,f), Z-ST (c,g), G-ST (d,h). Gradient tandem-mass spectrometry (gMS^2) of i) SD at m/z 843.9 (5+), j) E-ST at m/z 910.3 (5+), k) Z-ST at m/z 910.5 (5+) and l) G-ST at m/z 910.3 (5+) with different collision energy.

To detect the possible existence of overlapping isomers or conformers, especially the geometric isomers E-ST, Z-ST, and G-ST with the same molecular weight, TWIM-MS was utilized as the advanced level of MS analysis. In the TWIM-MS spectra (Figure 3e–h), the narrow drift time distribution for each charge state of these complexes was observed, indicative of the formation of a discrete structure as the sole assembly product. Furthermore, gradient tandem mass spectrometry (gMS^2) was employed to evaluate differences in the stability of double- and triple-decker complexes.^[24] All complexes SD, E-ST, Z-ST, and G-ST exhibited good stability, and fragment ion peaks were not observed below 8 V (Figure 3i–l). Notably, double-decker SD was more stable than triple-decker E-ST, Z-ST, and G-ST. Clear fragments of SD were observed until the voltage was up to 20 V, which was attributed to the relatively higher density of coordination sites.^[24]

Single crystals are the most suitable evidence to confirm the absolute configuration of molecules. However, all attempts to obtain single crystals of these complexes were unsuccessful, which were presumably attributed to the multicomponents of complicated structures as well as the long alkyl chains leading to unfavorable effects for intermolecular stacking.^[25] Fortunately, we substituted alkyl chains with tert-butyl group and obtained the single-crystal structure of E-STB, however, the other three single-crystal structures were still not acquired. Thus, density functional theory (DFT) calculations are performed to simulate the supramolecular structures of SD, Z-ST, and G-ST. In the single-crystal structure of triple-decker E-STB (Figure 4b), two claw-like tpy moieties, which were located on the opposite corner points of scaffold LHB, flipped to the bottom and coordinated with guest ligand trans-LG to form an “X-shaped”

propeller structure for preventing the formation of thermodynamically unstable oligomers of trans-LG.^[26] The distances between two layers of the C = C bond of TPE moieties were 5.1 and 5.1 Å, respectively, which were shorter than that of most of the reported supramolecular assemblies based on TPE.^[27] Compared with E-STB based on ditopic ligand, two TPE moieties of double-decker SD corresponding to host ligand LH and guest ligand tetra-LG almost completely overlapped with a distance of 5.4 Å (Figure 4a). As for Z-ST and G-ST, the convergence of geometric angles of ligands cis-LG and gem-LG facilitated the formation of “S-shaped” folded structures (Figure 4c,d). Three TPE moieties were stacked in parallel, and the distances between two layers of the C = C bond of TPE moieties were 5.2 and 5.6 Å for Z-ST and 5.4 Å for G-ST, respectively. Notably, this is the first example of a TPE-based metallar triple-decker structure to the best of our knowledge.

2.2. Photophysical Properties of Ligands and Complexes

After the structures were confirmed, the photophysical properties of ligands and complexes were investigated by UV–vis absorption and fluorescence spectroscopy (Figure 5a,b). All ligands exhibited a similar intense absorption band in the range of 250–400 nm. The absorption bands at ≈ 280 and 325 nm were corresponding to the π – π^* transition of the tpy groups and conjugated TPE core, while the band in the range of 350–400 nm was ascribed to the intramolecular charge transfer (ICT) absorption from an electron donor (TPE) to the electron acceptor (tpy) moieties.^[28] Fluorescence spectra of ligands LH, tetra-LG, and trans-/cis-/gem-LG revealed a weak emission

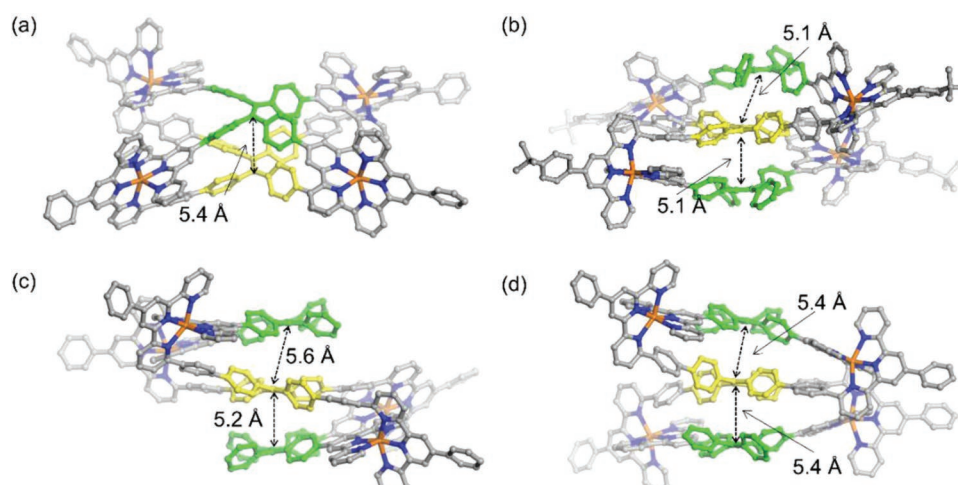


Figure 4. X-ray crystal structure of E-STB (b). Non-coordinated anions and solvent are omitted for clarity. DFT-optimized ground-state structures of SD (a), Z-ST (c), and G-ST (d). The hydrogen atoms and alkyl chains are omitted for clarity in the structures (C, gray or yellow; N, blue; Zn, orange).

centered at ≈ 400 and 525 nm in chloroform (absolute fluorescence quantum yield $\Phi_F < 0.1\%$), due to the non-radiative decay resulting from the free rotation of phenyl groups. Furthermore, the steric hindrance of phenyl groups also prevents the aggregation of tetra-LG which lead to weak emission. The emission at a longer wavelength was attributed to the ICT states. However, the host ligand LH exhibited enhanced emission in pure chloroform with Φ_F of 4.6% due to the bulkier size of the

substituents. In contrast, double-decker SD and triple-decker E-ST, Z-ST, and G-ST exhibited strong emissions even in dilute CH_3CN , with significantly improved Φ_F values of 9.1%, 4.5%, 2.9%, and 2.0%, respectively.

The time-resolved decayed lifetime (τ_s), radiative decay rates (k_r), and non-radiative decay rates (k_{nr}) of ligands and complexes were further investigated in CHCl_3 and CH_3CN , respectively (Figure 6, Tables S2 and S3, and Figures S58 and S59,

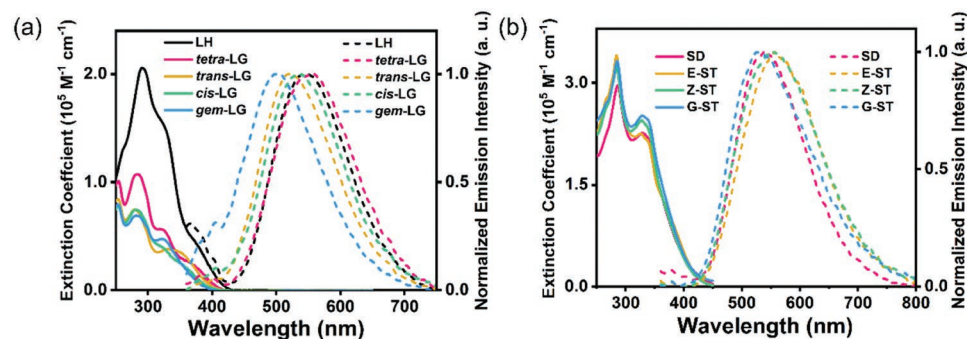


Figure 5. Absorption and fluorescent emission spectra of the a) ligands LH, tetra-LG, and trans-/cis-/gem-LG in CHCl_3 and b) complexes SD, E-ST, Z-ST, and G-ST in CH_3CN ($\lambda_{\text{ex}} = 340 \text{ nm}$, $c = 1.00 \mu\text{m}$).

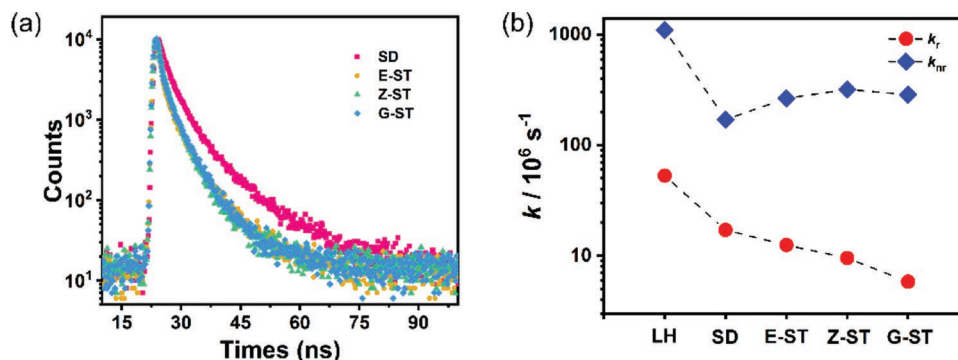


Figure 6. a) Fluorescence decays for complexes SD, E-ST, Z-ST, and G-ST in CH_3CN at 298 K. b) The radiative and non-radiative rate constants calculated from τ and Φ_F . k_r = radiative decay rate (Φ_F/τ_s); k_{nr} = non-radiative decay rate ($1/\tau_s - k_r$).

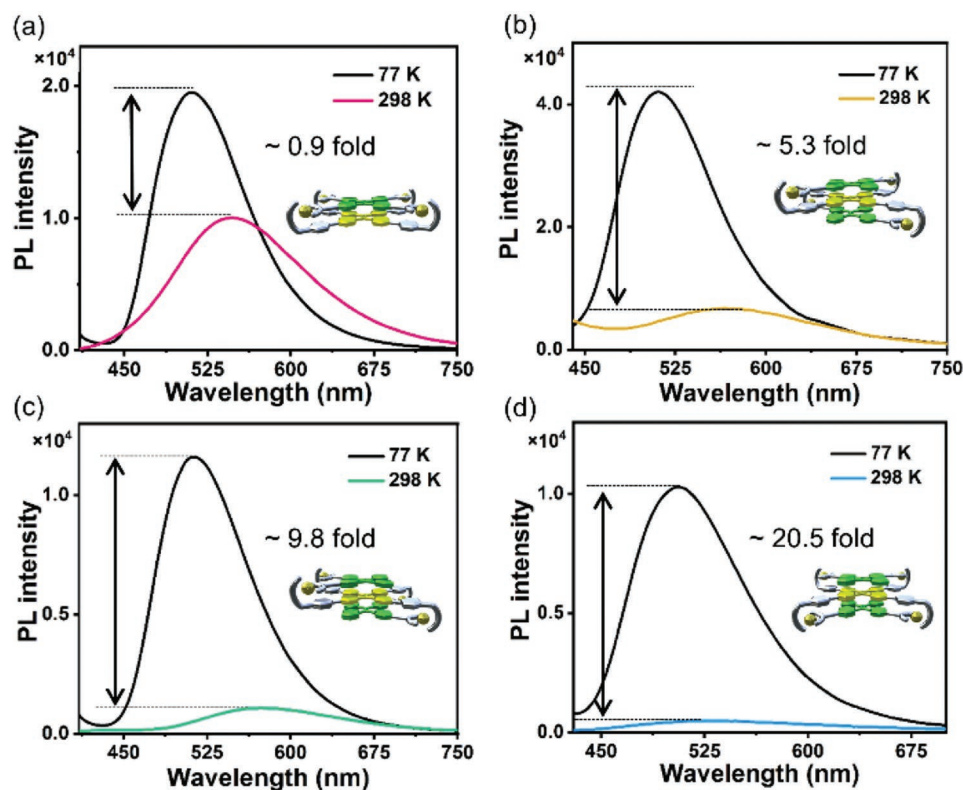


Figure 7. PL spectra of a) SD, b) E-ST, c) Z-ST, and d) G-ST in CH₃CN at 77 and 298 K (λ_{ex} = 365 nm, c = 1.00 μM).

Supporting Information). The τ_s of ligand LH in CHCl₃ was 0.87 ns. k_r and k_{nr} were calculated to be 5.29×10^7 and 1.10×10^9 s⁻¹, respectively. In contrast, k_{nr} of complexes SD, E-ST, Z-ST, and G-ST were 1.70×10^8 , 2.65×10^8 , 3.19×10^8 , and 2.86×10^8 s⁻¹, respectively, which were considerably less than that of ligands. Furthermore, k_{nr}/k_r of double-decker SD was 9.9, which was much smaller than that of host ligand LH (k_{nr}/k_r = 20.8). This result indicated that the double-decker SD with a rigid backbone might effectively suppress the energy loss via a non-radiative transition process by restricting the intramolecular rotation or vibration of TPE moieties.^[29] Interestingly, although the triple-decker assemblies E-ST, Z-ST, and G-ST comprise a higher number of TPE primitives, the phenyl groups of TPE without tpy substitution could slightly undergo rotation. Hence, these triple-decker complexes showed a weaker emission than that of double-decker SD. Especially for the geometric isomers of E-ST, Z-ST, and G-ST, although they bear the same number of TPE primitives, the k_{nr}/k_r ratio of G-ST was much higher than those of E-ST and Z-ST, which are related to the free rotation of the C = C bond for the gem-isomer in the excited states.^[30] In contrast, the C = C bonds of trans/cis-LG belonged to E-ST and Z-ST are hard to freely rotate in the excited states due to the restriction of the connecting moieties.^[31] It is worth mentioning that k_{nr}/k_r of triple-decker E-ST was smaller than that of triple-decker Z-ST, which might be attributed to the more restrictive chemical environment of “X-shaped” molecular conformation than “S-shaped” molecular conformation.

In addition, this result was verified by low-temperature fluorescence spectra of SD, E-ST, Z-ST, and G-ST (Figure 7).

The spectra were recorded by rapidly freezing the assemblies in CH₃CN to reach a glassy state with liquid nitrogen at 77 K.^[32] The emission intensities of the complexes at 77 K were improved in comparison with those at room temperature. However, owing to the different inhibition of intramolecular motion in the double- and triple-decker structures, the emission intensity of SD only increased by ≈ 0.9 -fold, while those of E-ST, Z-ST, and G-ST increased by ≈ 5.3 , 9.8, and 20.5-fold, respectively. The double- and triple-decker structures exhibited different increases in the emission intensity at 77 K, which were attributed to the different spatial arrangements. The intramolecular rotations and vibrations of TPE moieties in SD have been hindered considerably at room temperature, thereby leading to small changes in the molecular conformation compared with the conformation locked by liquid nitrogen at 77 K. However, triple-decker E-ST, Z-ST, and G-ST exhibited several freely rotatable phenyl groups at room temperature, which considerably contributed to the increase of the strong emission at 77 K.

Fluorescent properties of double-decker SD and triple-decker E-ST, Z-ST, and G-ST in a mixed solvent of CH₃CN as the good solvent and H₂O as the poor solvent were further investigated (Figure 8a–d). With increasing the water fraction from 0% to 90%, the fluorescence intensity of SD gradually increased and reached the maximum at 90%, with a Φ_F of 20.7% (Figure S60, Supporting Information). However, at the same concentration, triple-decker assemblies E-ST, Z-ST, and G-ST exhibited slightly different responses to the increase in the water fraction. The fluorescence intensity of E-ST, Z-ST, and G-ST become a slight increased when the water fraction was 10%. However,

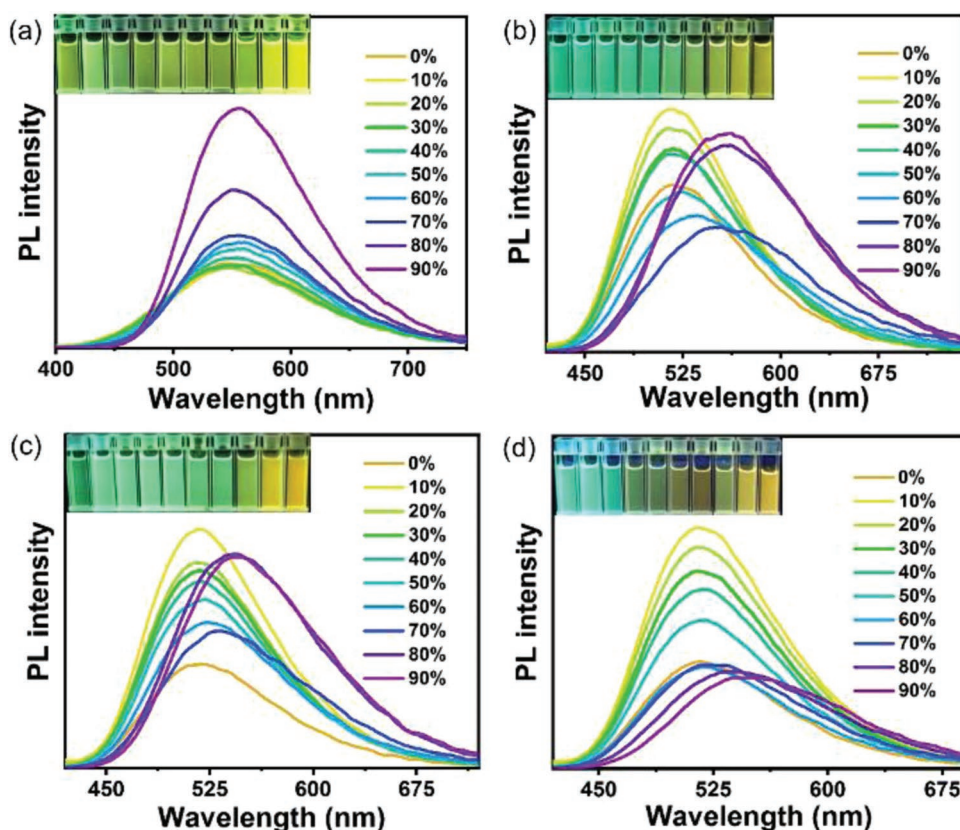


Figure 8. Fluorescence emission spectra of a) SD, b) E-ST, c) Z-ST, and d) G-ST versus the water fraction in the acetonitrile/water mixtures ($\lambda_{ex} = 340$ nm, $c = 1.00$ μ m). Inset: photographs of assemblies in acetonitrile/water mixtures with different fractions of water under excitation at 365 nm using an ultraviolet lamp at 298 K ($c = 1.00$ μ m).

the emission intensity decreased with the change in the water fraction from 20% to 60%, which was attributed to the enhanced intramolecular charge transfer process in polar solvents.^[25] With the further addition of water, the AIE phenomenon was observed, and emission intensities of triple-decker E-ST, Z-ST, and G-ST were enhanced, accompanying a large bathochromic shift (Figure 8b–d). Eventually, triple-decker E-ST, Z-ST, and G-ST exhibited yellow emission at a water fraction of 90%, with Φ_F values of 8.7%, 11.7%, and 20.3%, respectively. With the increase in the fraction of poor solvent, the rotation of phenyl rings was restricted by aggregation, leading to enhanced emission.

3. Conclusion

In summary, we reported the design and synthesis of a series of luminescent metallo-supramolecules with double- and triple-decker structures based on shape-complementary ligands. From ^1H NMR spectroscopy, COSY, ESI-MS, single-crystal analysis, and DFT calculations, the molecular conformation of SD was identified as a double-decker structure, Z-ST and G-ST were triple-decker “S-shaped” folded structures, while E-ST was a triple-decker “X-shaped” propeller structure. These special configurations imparted not only excellent emissions in dilute solution but also exhibited different fluorescent behaviors. This strategy is feasible and promising for constructing discrete

multi-decker architectures with precisely controlled orientations of chromophores. Therefore, compared with published supramolecular assemblies, this strategy is more general and can be applied to both polycyclic and nonpolycyclic aromatic hydrocarbons. And it would provide a new insight for studying the functions of dye aggregates and developing novel fluorescent molecules.

[CCDC 2153475, 2153476, and 2153478 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors are grateful for the support from the National Natural Science Foundation of China (22071079 for M.W.) and acknowledge staff members of the BL17B beamline of National Facility for Protein Science in Shanghai (NFPS) at Shanghai Synchrotron Radiation Facility for assistance during data collection, and Molecular Scale Lab for mass spectrometry characterization.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

multi-decker molecules, metallo-supramolecules, single crystals, terpyridine, tetraphenylethylene

Received: April 6, 2022
Revised: May 9, 2022
Published online: May 31, 2022

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